Re-analyzing the Effect of Right Heart Catheterization on ICU Mortality: A Doubly Robust Approach with Sensitivity Analysis

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June 12, 2026

1. Research Question & Dataset

Our primary causal question is:

> *What is the causal effect of right heart catheterization (RHC), administered during the initial intensive care unit (ICU) admission, on the mortality rate of critically ill adult patients?*

Right heart catheterization (RHC) is a minimally invasive procedure in which a catheter is threaded through a vein—typically in the neck, arm, or groin—into the right side of the heart. Widely considered the gold standard, it is used to directly measure blood pressure and oxygen levels within the heart and lungs. In a foundational study, Connors et al. (1996) investigated the causal effect of RHC on the 30-day mortality rate of critically ill ICU patients, employing propensity score matching to estimate the average treatment effect (ATE) and Rosenbaum bounds for sensitivity analysis. Given the significant advancements in causal inference estimators and sensitivity analyses over the past few decades, our team intends to re-examine this dataset using contemporary methodologies (e.g., AIPW, E-Values) to evaluate the robustness of the original findings.

Table 1: Summary of Analytical Approaches

Category	**Included Methods / Metrics**
Estimators	ATE, ATT, CATE
Methodologies	AIPW, IPW, Matching, Regression
Sensitivity Analysis	E-Values

The treatment is a binary variable denoting whether a patient received RHC. The primary outcome is 30-day mortality after RHC. We also consider 180-day mortality or length of hospital stay as secondary outcomes. The population is adult ICU patients in the SUPPORT RHC dataset, which consists of patients admitted to an ICU with one of several severe disease categories.

We will use the RHC dataset, derived from the SUPPORT observational study (Connors et al., 1996), which evaluates the efficacy of RHC in critically ill patients. The dataset includes n = 5,735 patients (2,184 treated; 38% of total observations) and utilizes a binary treatment indicator for RHC application within 24 hours, a binary outcome for 30-day survival, and 62 covariates constructed following Hirano and Imbens (2001) to adjust for confounding. The dataset is publicly available through ATbounds R package [1] and the website [2].

[1] https://search.r-project.org/CRAN/refmans/ATbounds/html/RHC.html

[2] https://hbiostat.org/data/repo/rhc.html

Table 2: Descriptive statistics Basic adjustment variables by RHC treatment group.

Variable	No RHC	RHC
Observations	3551	2184
Age	61.761 (0.290)	60.750 (0.334)
Years of education	11.569 (0.053)	11.856 (0.068)
Female	0.461 (0.008)	0.415 (0.011)
Black race	0.165 (0.006)	0.153 (0.008)
Other race	0.060 (0.004)	0.065 (0.005)
Income \$11k-\$25k	0.201 (0.007)	0.207 (0.009)
Income \$25k-\$50k	0.141 (0.006)	0.180 (0.008)
Income >\$50k	0.072 (0.004)	0.089 (0.006)
Cardiovascular symptoms	0.160 (0.006)	0.204 (0.009)
Congestive heart failure	0.168 (0.006)	0.195 (0.008)
Myocardial infarction	0.030 (0.003)	0.043 (0.004)
APACHE score	50.934 (0.316)	60.739 (0.434)
PaO2/FIO2 ratio	240.627 (1.958)	192.433 (2.258)
Mean blood pressure	84.869 (0.652)	68.198 (0.733)
Glasgow coma score	22.253 (0.526)	18.973 (0.605)
DNR on day 1	0.141 (0.006)	0.071 (0.005)
Death (30D)	0.306 (0.008)	0.380 (0.010)
Death (180D)	0.630 (0.008)	0.680 (0.010)

NOTE: cells report mean or proportion with standard error in parentheses.

Table 2 shows the descriptive statistics of our dataset. An evidence of confounding could be observed that there are larger differences between groups on physiology variables (e.g. blood pressure, APACHE score, etc). The crude mortality rates were also higher in the RHC group.

2. Causal Model

Figure 1 shows our preliminary assumptions. RHC is not assigned randomly. Doctors are more likely to use it for patients with severe condition related to heart disease or respiratory instability.

Figure 2 summarizes the detailed causal relationships we developed from preliminary DAG from dataset, between important representative variables and our variables of interest (treatment, outcome).

We give justifications for our proposed causal directions below:

Age → {Chfhx, Aps1, Dnr1}: Age is already encoded into the APACHE III score (7). Furthermore, epidemiological literature establishes age as a primary predictor for the Do Not Resuscitate (DNR) orders upon ICU admission, independent of acute physiological derangement (1; 6).

Cat1 → {Meanbp1, Pafi1, Scoma1, Swang1}: The original paper's methodology notes that these diagnostic categories act as direct clinical triggers for Pulmonary Artery Catheter (PAC) insertion based on standard ICU protocols (2).

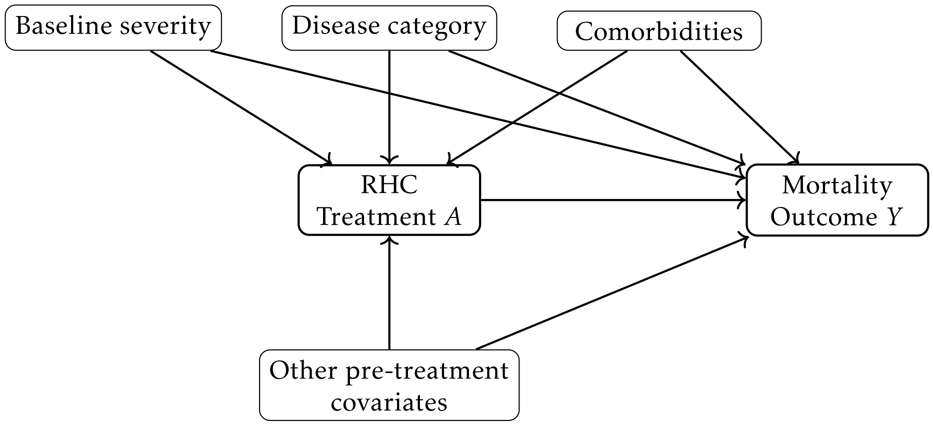


Figure 1: Preliminary causal graph for RHC dataset.

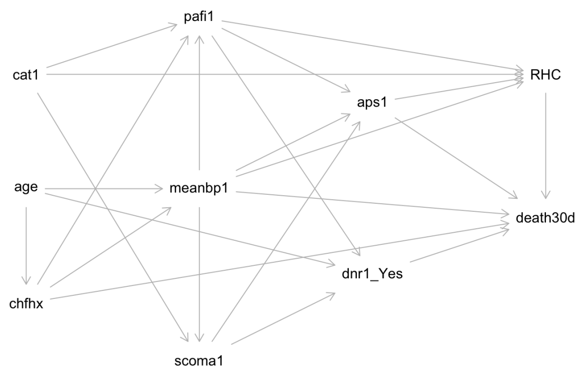


Figure 2: Proposed DAG for the relationship between relevant variables in RHC study

Chfhx → {Meanbp1, Pafi1}: Because a failing heart cannot pump effectively under stress, these patients are highly vulnerable to sudden crashes. This directly causes dangerously low blood pressure (Meanbp1) and fluid backing up into the lungs, which blocks oxygen from entering the blood (Pafi1) (4).

Meanbp1 → {Scoma1}: Severe shock causes a massive drop in blood pressure. Without enough pressure to pump blood to the brain, the patient suffers acute brain dysfunction, which shows up clinically as a lowered Glasgow Coma Score (Scoma1) (10).

Meanbp1, Pafi1, Scoma1 → {Aps1, Death}: These abnormal vital signs are used directly to calculate the APACHE III severity score (7). The worse a patient's blood pressure and oxygen levels are, the higher their risk of death.

Scoma1 → {Dnr1}: When a patient becomes severely unresponsive or enters a deep coma, it usually triggers an urgent meeting with the family. These discussions often lead to establishing a Do Not Resuscitate (DNR) order or withdrawing life support (9).

Meanbp1, Pafi1 → {Swang1}: Severe drops in blood pressure and complex oxygen issues are the primary reasons doctors insert this catheter. It helps them figure out if lung failure is caused by the heart or something else, and guides exactly how to dose blood pressure medications (5; 8).

Table 3: Node Definitions and Keys

Node	Key
`RHC`	Right Heart Catheterization (RHC) status during the first 24 hours (Treatment, A).
`death30d`	All-cause mortality up to 30 days (Outcome, Y).
`age`	Patient age in years.
`cat1`	Primary disease category prompting ICU admission (**Category:** CHF, Cirrhosis, Colon Cancer, Coma, COPD, Lung Cancer, MOST w/ Malignancy, MOSF w/ Malignancy, MOSF w/sepsis).
`chfhx`	Baseline history of Congestive Heart Failure.
`meanbp1`	Mean blood pressure recorded on day 1.
`pafi1`	Ratio of arterial oxygen partial pressure to fractional inspired oxygen (PaO_2/FIO_2) on day 1.
`scoma1`	Glasgow Coma Score on day 1.
`aps1`	APACHE III score on day 1.
`dnr1_yes`	Do Not Resuscitate (DNR) status established on day 1.

Aps1 $\rightarrow$ {Swang1}: The sicker a patient is overall, the more likely doctors are to use aggressive, invasive tools to monitor them, regardless of any single vital sign (2).

Dnr1 $\rightarrow$ {Swang1, Death}: A DNR order prevents the medical team from performing highly invasive procedures like inserting this catheter (3). Having a DNR is also a predictor that the patient is likely to pass away soon.

3. Identification

3.1 Estimands

The target estimands are the average treatment effect,

$$\text{ATE} = E\{Y(1) - Y(0)\},$$

and the average treatment effect among treated patients,

$$\text{ATT} = E\{Y(1) - Y(0) \mid A = 1\}.$$

Let Z denote the measured baseline adjustment set. We use the backdoor adjustment criterion. The required assumptions are consistency, positivity, and conditional exchangeability:

$$Y = Y(A), \qquad 0 < P(A = 1 \mid Z) < 1, \qquad Y(a) \perp A \mid Z.$$

Under these assumptions, the ATE is identified by

$$E\{Y(1) - Y(0)\} = E\left[E(Y \mid A = 1, Z) - E(Y \mid A = 0, Z)\right].$$

The ATT is identified by averaging the same conditional contrast over the treated covariate distribution:

$$E\{Y(1) - Y(0) \mid A = 1\} = E_{Z|A=1}\left[E(Y \mid A = 1, Z) - E(Y \mid A = 0, Z)\right].$$

3.2 Adjustment Sets

Table 4 shows the basic and detailed variables in the datasets. We then defined three nested sets of adjustment covariates. The basic set adds key clinical history and day-1 physiology variables and is our primary adjustment set. The detailed set adds additional histories, labs, diagnosis indicators, and physician-predicted survival variables. The basic set is the main analysis because it captures the primary confounding pathways while avoiding the instability that can come from a very high-dimensional propensity model. The basic set is our primary specification because it captures the main measured backdoor pathways, especially confounding by acute illness severity, while avoiding the instability that can arise from a very high-dimensional propensity model.

Table 4: Categorization of Baseline Variables

Category	Variables
Demographics (Basic)	`age`, `edu`, `sex_Female`, `race_black`, `race_other`, `income1`, `income2`, `income3`
Demographics (Detailed)	`ninsclas_Medicaid`, `ninsclas_Medicare`, `ninsclas_Medicare_and_Medicaid`, `ninsclas_No_insurance`, `ninsclas_Private_and_Medicare`
Clinical History (Basic)	`cardiohx`, `chfhx`, `amihx`
Clinical History (Detailed)	`dementhx`, `psychhx`, `chrpulhx`, `renalhx`, `liverhx`, `gibledhx`, `malighx`, `immunhx`, `transhx`
Day 1 Physiology (Basic)	`aps1`, `pafi1`, `meanbp1`, `scoma1`, `dnr1_Yes`
Day 1 Physiology (Detailed)	`resp1`, `hrt1`, `paco21`, `ph1`, `wblc1`, `hema1`, `sod1`, `pot1`, `crea1`, `bili1`, `alb1`, `temp1`, `wtkilo1`, `wt0`
Diagnosis Indicators	`cat1_CHF`, `cat1_Cirrhosis`, `cat1_Colon_Cancer`, `cat1_Coma`, `cat1_COPD`, `cat1_Lung_Cancer`, `cat1_MOSF_Malignancy`, `cat1_MOSF_Sepsis`, `cat2_Cirrhosis`, `cat2_Colon_Cancer`, `cat2_Coma`, `cat2_Lung_Cancer`, `cat2_MOSF_Malignancy`, `cat2_MOSF_Sepsis`, `resp_Yes`, `card_Yes`, `neuro_Yes`, `gastr_Yes`, `renal_Yes`, `meta_Yes`, `hema_Yes`, `seps_Yes`, `trauma_Yes`, `ortho_Yes`, `ca_Metastatic`, `ca_Yes`
Other Variables	`surv2md1`, `das2d3pc`

4. Estimation

The primary estimand is the ATE of RHC on 30-day mortality, reported on the risk-difference scale. This captures the absolute change in mortality probability. We also report the ATT as a secondary estimand because it answers a related question: among severe patients who actually received RHC, how would their mortality change?

All primary estimators use the basic adjustment set defined in Section 3. The minimal and detailed adjustment sets are reserved for the adjustment-set sensitivity analysis in Appendix A.1.

4.1 AIPW and IPW Estimators

Our preferred estimator is augmented inverse probability weighting (AIPW). AIPW combines an outcome regression model for $E(Y \mid A, Z)$ with a propensity-score model for $P(A = 1 \mid Z)$. For the ATE, the estimator averages the augmented contrast

$$\{\hat{\mu}_1(Z) - \hat{\mu}_0(Z)\} + \frac{A\{Y - \hat{\mu}_1(Z)\}}{\hat{e}(Z)} - \frac{(1-A)\{Y - \hat{\mu}_0(Z)\}}{1 - \hat{e}(Z)},$$

where $\hat{\mu}_a(Z)$ is the fitted outcome regression under treatment level a, and $\hat{e}(Z)$ is the estimated propensity score. The appeal of AIPW is double robustness: under the identification assumptions, it remains consistent if either the outcome model or the propensity model is correctly specified.

We also report Hájek IPW estimates. IPW uses the propensity score to reweight patients so that the weighted treated and untreated groups resemble the target population with respect to measured covariates. The Hájek normalization stabilizes the weights by dividing by the sum of weights within each treatment arm. We use IPW mainly as a transparent weighting-based comparison to AIPW. For both IPW and AIPW, standard errors and confidence intervals are obtained with 200 bootstrap resamples.

Regression adjustment and propensity-score matching are included in the result tables as informal specification checks, but they are not the focus of our estimation strategy.

Table 5: ATE by different causal estimators

Estimand	Method	Estimate	SE	95%CI
	Naive comparison	0.0736	0.0130	[0.0483, 0.0990]
ATE	Regression adjustment	0.0614	0.0122	[0.0374, 0.0853]
ATE	PS matching	0.0515	0.0164	[0.0194, 0.0835]
ATE	Hajek IPW	0.0564	0.0128	[0.0314, 0.0814]
ATE	AIPW	0.0565	0.0125	[0.0320, 0.0810]

Note: The SE for IPW, AIPW were obtained with bootstrapping with 200 times.

All adjusted ATE estimates are positive. The AIPW ATE is 0.0565 with a 95% CI of [0.0320, 0.0810], meaning that RHC is estimated to increase 30-day mortality by about 5.7 percentage points under the measured-confounding adjustment. The ATT estimates are slightly smaller; the

Table 6: ATT by different causal estimators

Estimand	Method	Estimate	SE	95%CI
ATT	Regression adjustment	0.0614	0.0122	[0.0374, 0.0853]
ATT	PS matching	0.0407	0.0169	[0.0075, 0.0739]
ATT	Hajek IPW	0.0468	0.0140	[0.0194, 0.0741]
ATT	AIPW	0.0487	0.0135	[0.0222, 0.0752]

Note: The SE for IPW, AIPW were obtained with bootstrapping with 200 times.

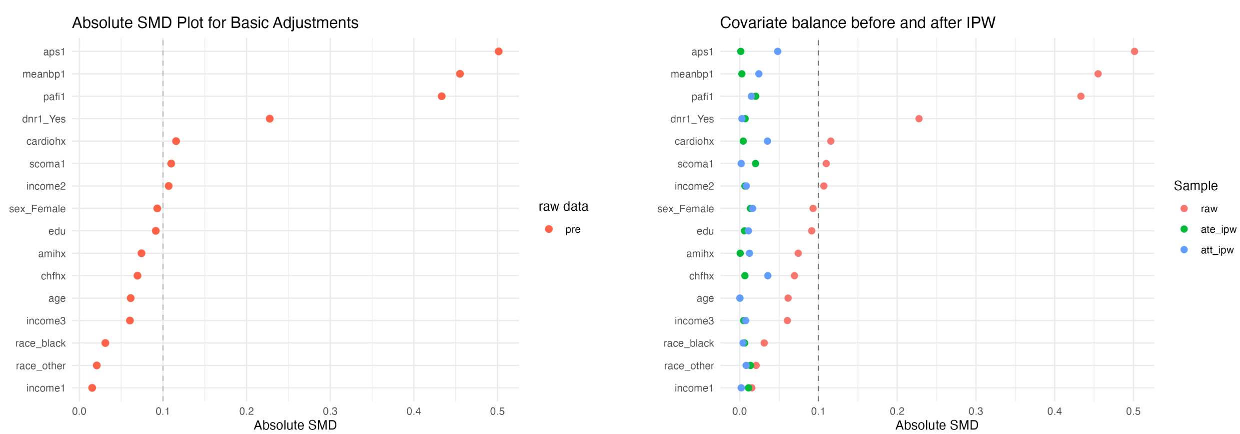
AIPW ATT is 0.0487 with a 95% CI of [0.0222, 0.0752].

5. Diagnostics

5.1 Covariate Balance

The key diagnostic for the IPW and AIPW analyses is whether the propensity-score model adequately balances the measured confounders. Figure 3 shows large pre-adjustment standardized mean differences for acute severity variables, especially APACHE III score, mean blood pressure, PaO2/FIO2 ratio, and DNR status. These imbalances are exactly what we would expect under confounding by indication: patients who received RHC were clinically sicker at baseline.

After weighting, the standardized mean differences are mostly close to zero. This does not prove conditional exchangeability, because unmeasured confounding remains possible, but it supports the claim that the measured components of the basic adjustment set are substantially better balanced in the weighted pseudo-population.



(a) Before adjustment

(b) After IPW

Figure 3: Standardized mean differences for the basic adjustment variables.

5.2 Positivity and Overlap

The second core diagnostic is overlap in the propensity score. Positivity requires that patients with similar baseline covariates have a nonzero probability of receiving either treatment level.

Figure 4 shows that overlap is imperfect, but not catastrophic. The treated group tends to have higher propensity scores, again reflecting clinical selection into RHC, but there is still substantial common support.

The effective sample size under the primary Hájek IPW analysis is about 4,411 for the ATE and 4,020 for the ATT. This indicates that weighting reduces some data relative to the original sample of 5,735 patients, but the estimate is not dominated by only a small number of extreme-weight observations.

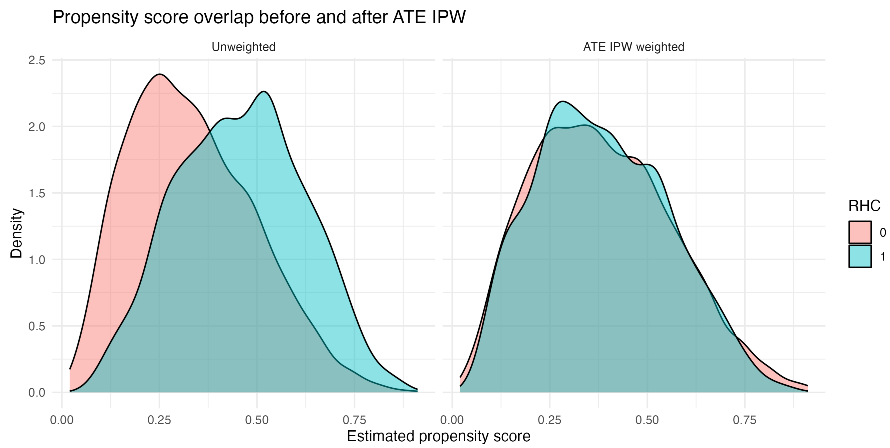


Figure 4: Propensity-score overlap under the basic adjustment set.

The remaining assumptions are not directly testable. Consistency depends on treating "RHC" as a sufficiently well-defined intervention in this clinical setting, and conditional exchangeability depends on whether the measured baseline covariates capture the major sources of treatment selection. These assumptions will be examined in the sensitivity analysis and interpretation section.

6. Results & Interpretation

6.1 Primary Finding

Our result suggests that RHC is still estimated to increase 30-day mortality. The AIPW ATE estimate is 0.0565 with a 95% CI of [0.0320, 0.0810]. Interpreted on the risk-difference scale, this corresponds to about a 5.7 percentage-point increase in 30-day mortality if all patients in the target ICU cohort received RHC rather than not receiving RHC.

6.2 Comparison Across Estimators

Figure 5 summarizes the main risk-difference estimates. The naive difference is largest, about 7.4 percentage points, which is expected because patients receiving RHC were initially sicker. After adjustment, the estimates shrink but remain positive. Across regression adjustment, matching, IPW, and AIPW, the adjusted ATE ranges from about 5.1 to 6.1 percentage points, and the adjusted ATT ranges from about 4.1 to 6.1 percentage points.

The similarity between the IPW and AIPW estimates is reassuring because the two estimators use the propensity score differently. IPW relies directly on weighting, while AIPW adds an outcome model correction. This suggests that the main result is not driven by a single estimator implementation.

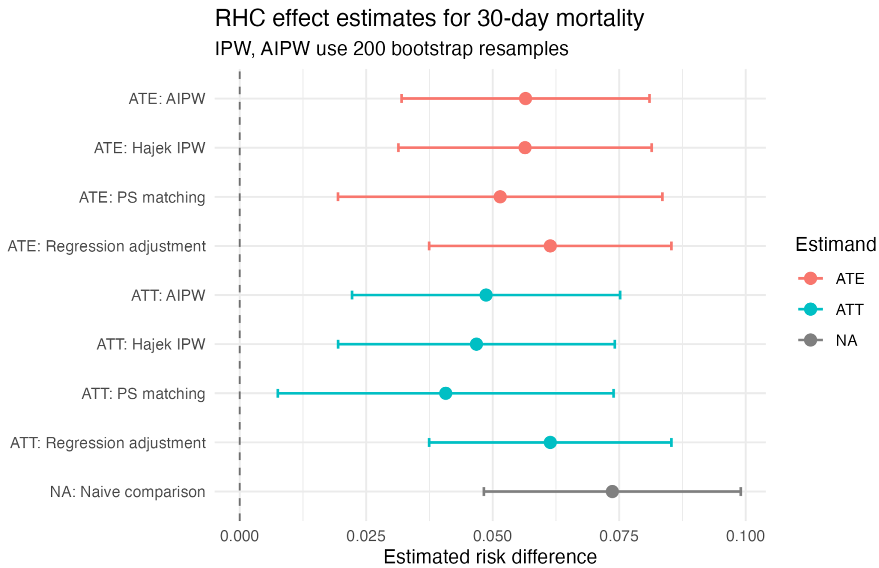


Figure 5: Estimated effect of RHC on 30-day mortality across estimators.

6.3 Risk-Ratio Interpretation

On the risk-ratio scale, the adjusted estimates also suggest higher mortality among RHC patients. The AIPW ATE risk ratio is 1.18, and the AIPW ATT risk ratio is 1.15.

Table 7: Risk Ratio Results by different causal estimators

Estimand	Method	Estimate	SE	95%CI
Naive	Naive comparison	1.2404	0.0461	[1.1532, 1.3342]
ATE	Regression adjustment	1.1931	0.0425	[1.1128, 1.2793]
ATE	PS matching	1.1614	0.0300	[1.1040, 1.2218]
ATE	Hajek IPW	1.1780	0.0430	[1.0967, 1.2654]
ATE	AIPW	1.1789	0.0422	[1.0990, 1.2646]
ATT	Regression adjustment	1.1889	0.0442	[1.1053, 1.2788]
ATT	PS matching	1.1200	0.0453	[1.0346, 1.2124]
ATT	Hajek IPW	1.1404	0.0451	[1.0554, 1.2322]
ATT	AIPW	1.1470	0.0439	[1.0641, 1.2363]

Note: IPW, and AIPW SEs were obtained with 200 bootstrap resamples on the log-risk-ratio scale.

6.4 Exploratory Subgroup CATE

Subgroup CATE analysis suggests that the estimated risk difference is generally larger in clinically severe subgroups. The AIPW ATE is about 0.105 among high-APACHE patients, 0.089 among patients age 65 or older, and 0.206 among patients with day-1 DNR status. However, these estimates would be unstable and should not be treated as definite estimation since the sample sizes are smaller.

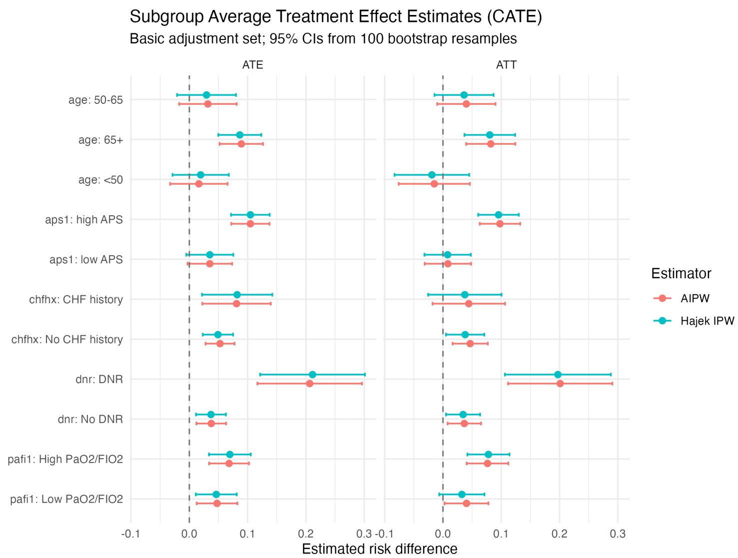


Figure 6: Exploratory CATE estimates for 30-day mortality.

6.5 Sensitivity to Unmeasured Confounding

The main limitation is unmeasured confounding. For the AIPW ATE risk ratio of 1.18, the E-value is about 1.64, with a confidence-limit E-value about 1.44. This means that an unmeasured confounder associated with both RHC and 30-day mortality by risk ratios of roughly 1.64 each, beyond measured covariates, could explain away the point estimate. Table 9 shows that the strongest observed benchmark covariate, day-1 DNR status, has a weaker-leg risk ratio of about 1.98, larger than the E-value. Thus, the estimated harmful association is not robust to plausible unmeasured confounding of similar strength.

Table 8: Evalue and Risk Ratio

Estimator	Risk (RHC)	Risk (No RHC)	RR	SE	95% CI	E-value	E-value CI
AIPW	0.3723	0.3158	1.1789	0.0400	[1.1029, 1.2500]	1.6381	1.4397
Regression adjustment	0.3707	0.3115	1.1901	0.0418	[1.1192, 1.2746]	1.6657	1.4846

Note: SEs were obtained with 100 bootstrap resamples on the risk-ratio scale.

Table 9: Covariate Risk Ratio Benchmark

covariate	RR_AW	RR_AY	weaker_leg
dnr1_Yes	1.9800	2.0278	1.9800
aps1	1.4900	1.7179	1.4900
meanbp1	1.5450	1.3137	1.3137
cardiohx	1.2789	1.2817	1.2789
income2	1.2780	1.1813	1.1813
amihx	1.4711	1.1797	1.1797
chfhx	1.1594	1.3163	1.1594
pafi1	1.4425	1.1474	1.1474
income3	1.2273	1.1273	1.1273
scoma1	1.0629	1.6184	1.0629
age	1.0595	1.1974	1.0595
race_other	1.0839	1.0475	1.0475
income1	1.0307	1.0630	1.0307
sex_Female	1.1113	1.0181	1.0181
race_black	1.0740	1.0105	1.0105
edu	1.1817	1.0081	1.0081

6.6 Conclusions

In summary, under measured-confounding adjustment, RHC is estimated to increase 30-day mortality by roughly 5 percentage points. The conclusion is consistent across several estimators and diagnostic checks, but it is causally fragile because unmeasured severity, physician judgment, hospital fixed effect, or treatment-limitation factors could plausibly be strong enough to explain the association.

7. Broader Context

In a broader context, there is more room to improve the estimation of the RHC effect on the mortality rate. Specifically, we were unable to use Difference-in-Differences (DID) to estimate the average treatment effect on the treated (ATT) because the RHC dataset lacks sufficient pre- and post-treatment information. Although it includes the baseline conditions of critically ill patients prior to treatment assignment, the 30-day observation period following treatment is so brief that long-term outcomes caused by the intervention remain unknown. Additionally, a valid instrumental variable could not be identified among the variables in the RHC dataset. Because the use of right heart catheterization was primarily determined by patient severity in the ICU, unobserved confounders correlated with the treatment are extremely difficult to measure. If we have the opportunity to consult a cardiovascular surgeon specializing in right heart catheterization, discussing causal discovery would be a valuable extension of this work. Even though both the DAG we generated and the one in the original paper were constructed with sufficient clinical references, conducting advanced causal discovery with domain experts could yield a DAG that is more representative of the true relationships between these variables.

Appendix: Additional Sensitivity Checks

A.1 Sensitivity to Adjustment-Set Choice

The adjustment-set sensitivity analysis suggests that demographic-only adjustment is insufficient: the minimal-set AIPW ATE is about 0.081, while the basic-set estimate is about 0.056. Moving from the basic to the detailed set changes the ATE only modestly, which supports using the basic set as the primary adjustment set. The ATT is somewhat more sensitive to the detailed set, our guess is that ATT uses less samples, hence it is more unstable for higher dimensional case.

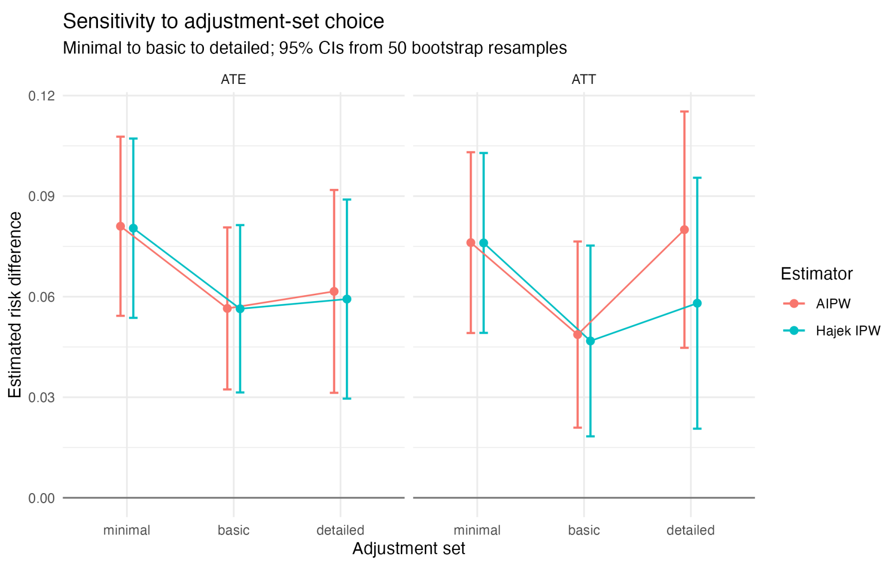


Figure 7: Sensitivity to the minimal, basic, and detailed adjustment sets.

A.2 AIPW Partial-R^2 Sensitivity

As a secondary sensitivity check, we used a cross-fitted AIPW partial-R^2 analysis with sensemakr-DML. The ATE estimate is 0.0560 with a 95% CI of [0.0301, 0.0819], similar to the main AIPW estimate. The robustness value for the point estimate is about 0.055, suggesting that an omitted factor explaining a 5.5 percent of both treatment selection and mortality could still change the causal interpretation. Therefore, it is not plausibly robust.

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